

# Phonon Pulse Templates for SuperCDMS SNOLAB Run 4

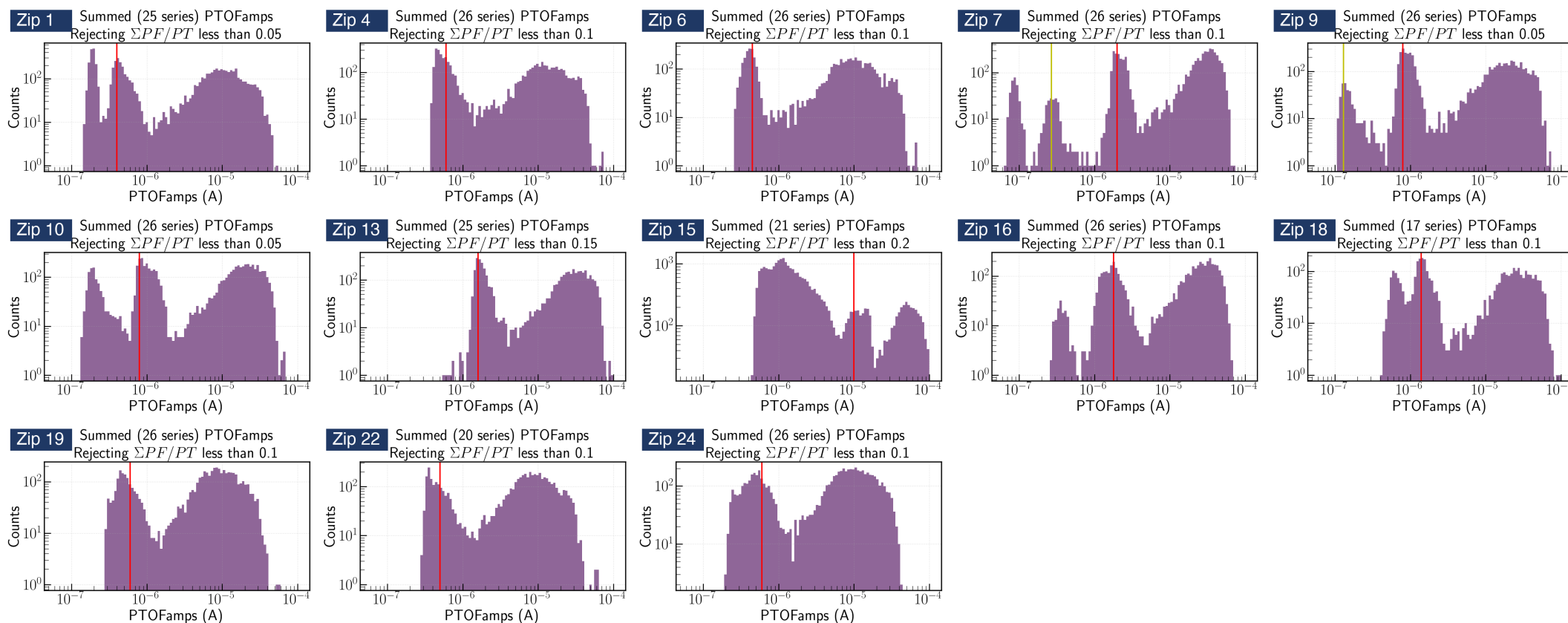
K-line event selection · free-pretrigger two-exponential fits · 1x1 and NxM (PCA) templates

Zhiheng Li — July 15, 2026

# From the Ge-activation K-line to an event sample

Goal: one clean, minimally-biased pulse population per detector × channel

- Per detector, the ops-shift study marked the 10.37 keV K-line position in PTOFamps — the red line in each panel below
- Our event selection: a PTOFamps window bracketing the red line (position  $\times/\div 1.35$ ) — deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata
- 13 detectors (zips) × 27–30 series → ~120 GB raw cache; every later cut stays explicit and reversible

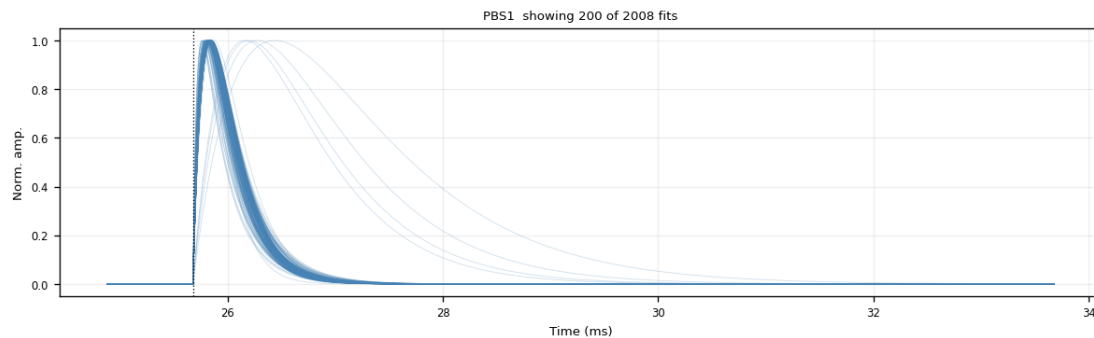


All 13 detectors, PTOFamps spectra (ops-shift study): red line = fitted mean of the 10.37 keV K-line. Selection window per detector = red-line position  $\times/\div 1.35$ .

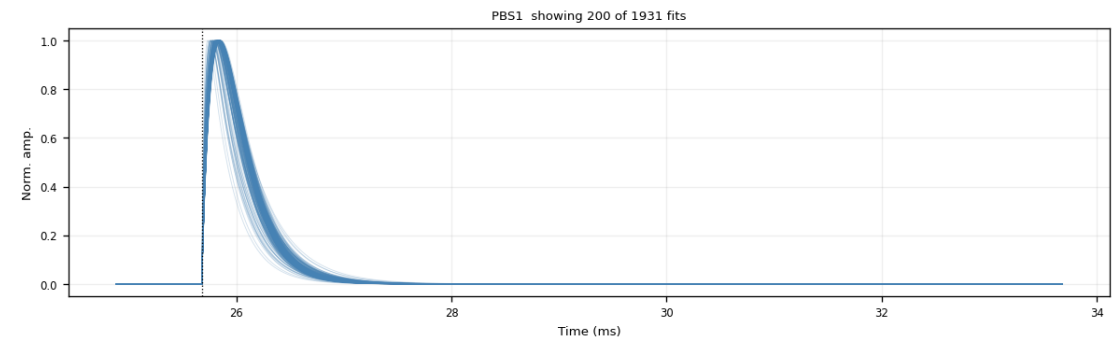
# Per-trace algorithm: low-pass → free-pretrigger fit → align

- 1. 100 kHz 4th-order Butterworth low-pass (steady-state init — no filter start-up transient)
- 2. Baseline = median of samples 2000–12000, subtracted; normalize by the global trace peak
- 3. Two-exponential fit — all 5 parameters free, including the onset:
  - $y(t) = A \cdot [\exp(-(t-t_0)/\tau_{\text{fall}}) - \exp(-(t-t_0)/\tau_{\text{rise}})] + b$ ,  $t_0 \in 16050 \pm 3000$  samples
  - fit window 12550–24050 (stride 4);  $\tau_{\text{rise}} \in [1 \mu\text{s}, 5 \text{ ms}]$ ,  $\tau_{\text{fall}} \in [10 \mu\text{s}, 20 \text{ ms}]$
- 4. Quality numbers —  $\text{fit\_ok} := A > 0$  and  $0 < \tau_{\text{rise}} < \tau_{\text{fall}}$ ;  $\text{NRMSE} := \text{RMS}(\text{residual})/\text{fit peak}$  (recorded, no cut yet)
- 5. Align — shift the measured trace by (fitted onset – 16050), sub-sample interpolation: a pure translation

• Why a free onset? Real trigger times vary event-by-event — fitted onsets cluster  $\approx 230$  samples after the nominal 16050. Pinning the onset distorts every other parameter.



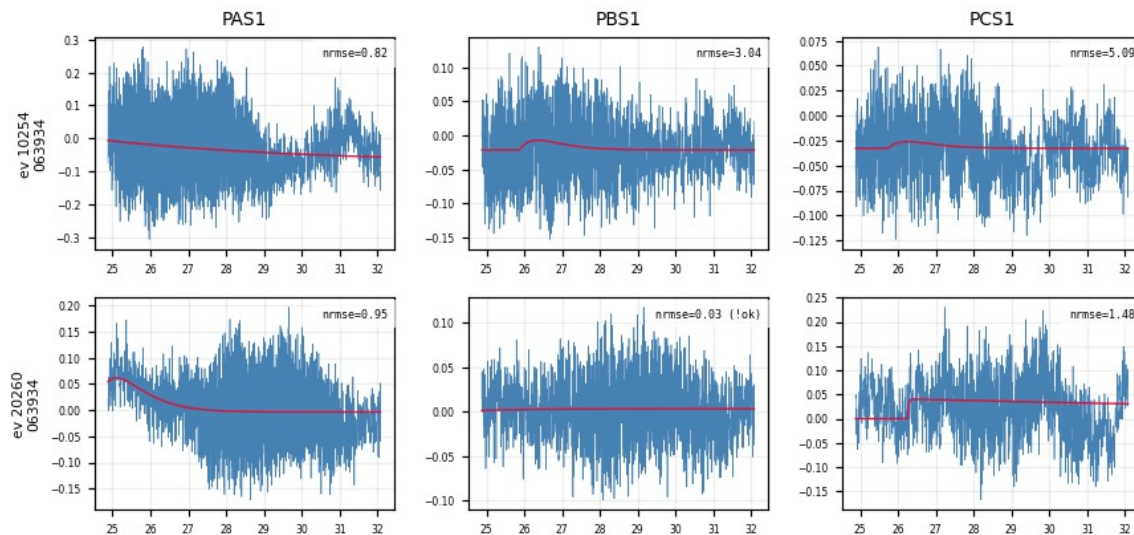
*Fitted curves at common onset, peak-normalized — ALL  $\text{fit\_ok}$  events (no cut)*



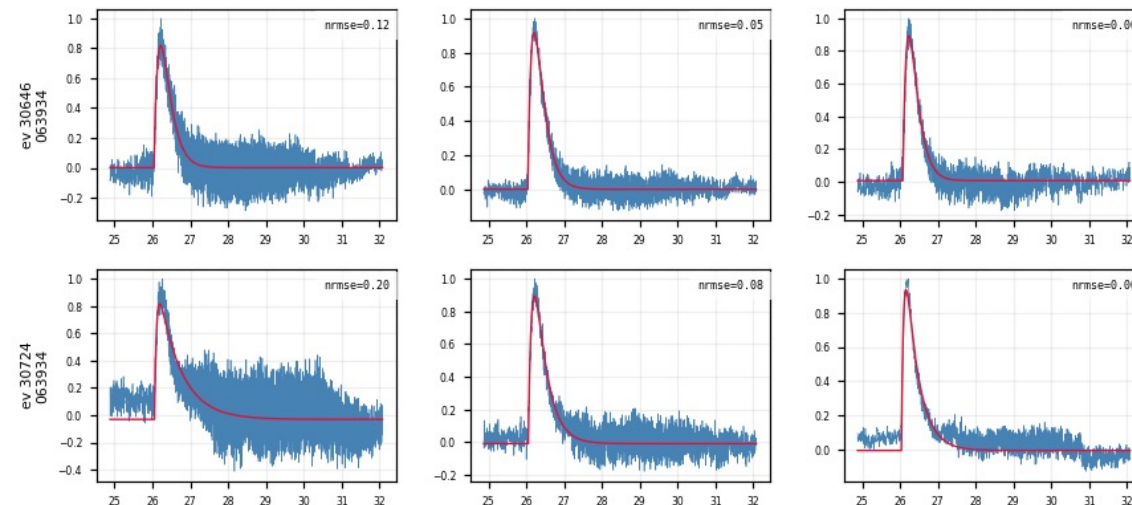
*Same, after  $\text{NRMSE} \leq 0.4$  — a single tight shape family remains (Z7 PBS1)*

# Fit quality at a glance — event × channel grids

- One row per event, one column per channel: low-passed trace (blue) vs 2-exp fit (red), NRMSE stamped per panel
- The same event fits consistently across channels; a noise trigger fails in all channels at once
- Every figure carries its full processing chain in the header → self-documenting



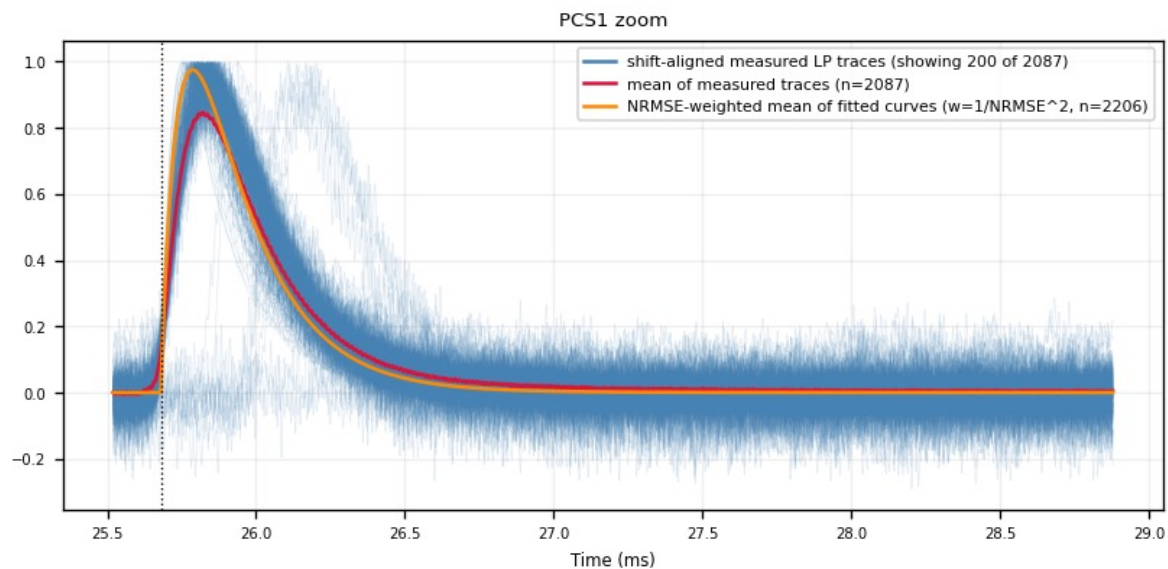
Noise triggers: the fit fails in every channel at once



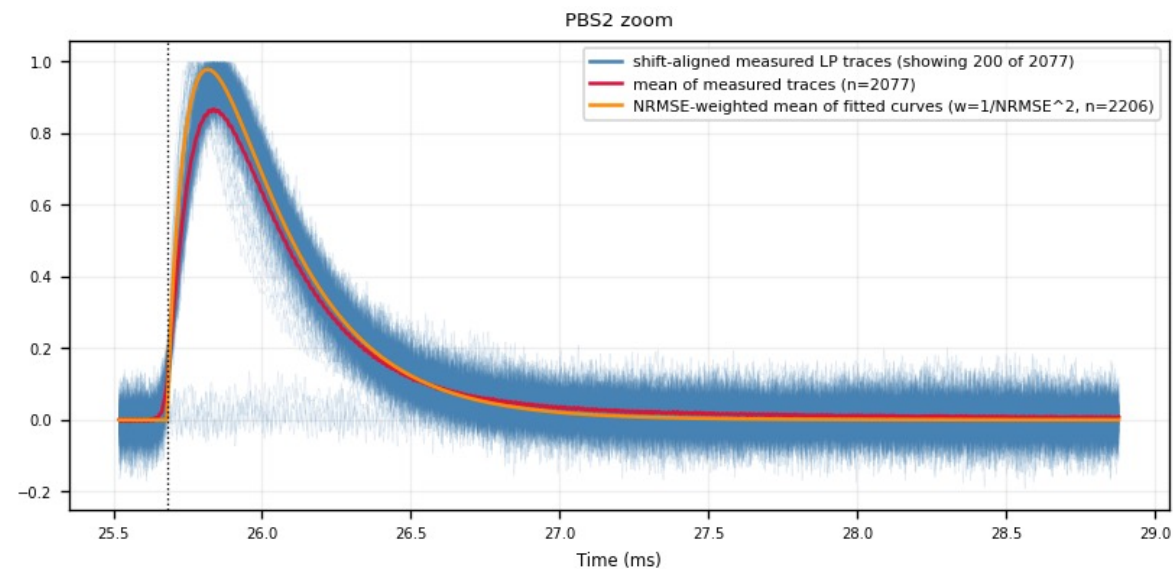
K-line events: consistent good fits across channels

# Alignment result — overlaid measured traces

- All fit\_ok traces shifted to the common onset (blue) + point-by-point mean (red) + NRMSE-weighted mean of the fitted curves (orange,  $w = 1/\max(\text{NRMSE}, 0.01)^2$ )
- Peak zoom on a quiet detector: a tight bundle — the template input is well defined. Note the faint displaced bundle beside the main one, visible on both crystal faces (S1 and S2) — more on it later



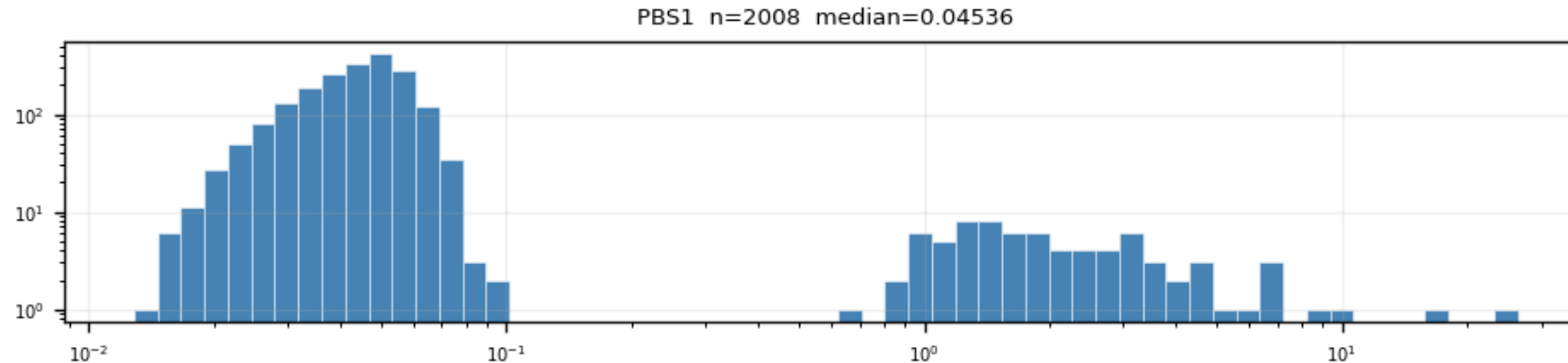
Z7 PCS1 (S1 face) — peak zoom; the faint displaced bundle is clearly visible



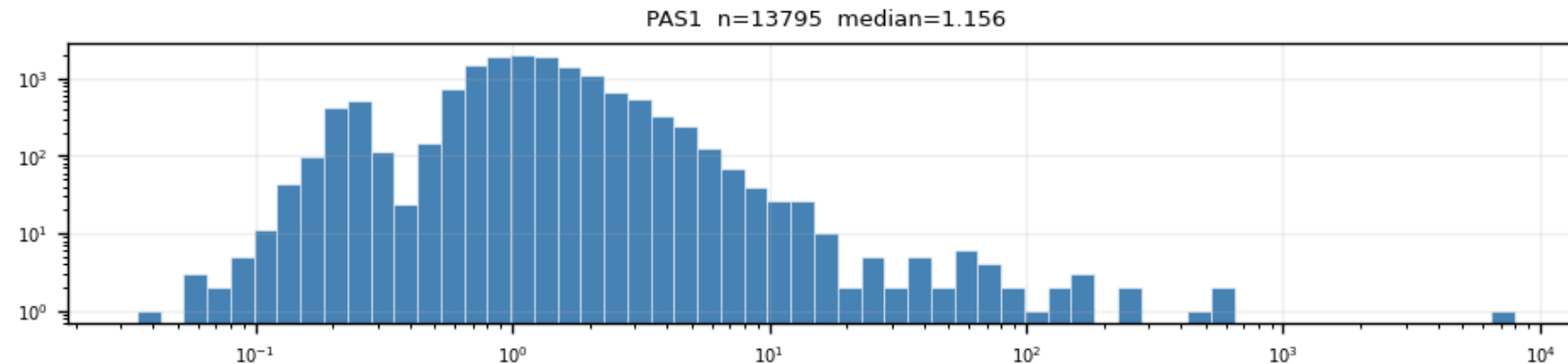
Z7 PBS2 (S2 face) — same faint displaced bundle

# Quality cut 1: $\text{NRMSE} \leq 0.4$ — where the number comes from

- NRMSE of fit\_ok events is bimodal: good fits (median  $\approx 0.05$ – $0.1$ ) vs noise triggers ( $\approx 1$ – $2$ ), valley at  $\approx 0.4$ – $0.5$
- The cut sits in the valley — it is read off the distribution, not tuned on the templates
- Weak detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): the PTOF window admits a noise-dominated mixture — this cut is what extracts the real-pulse population



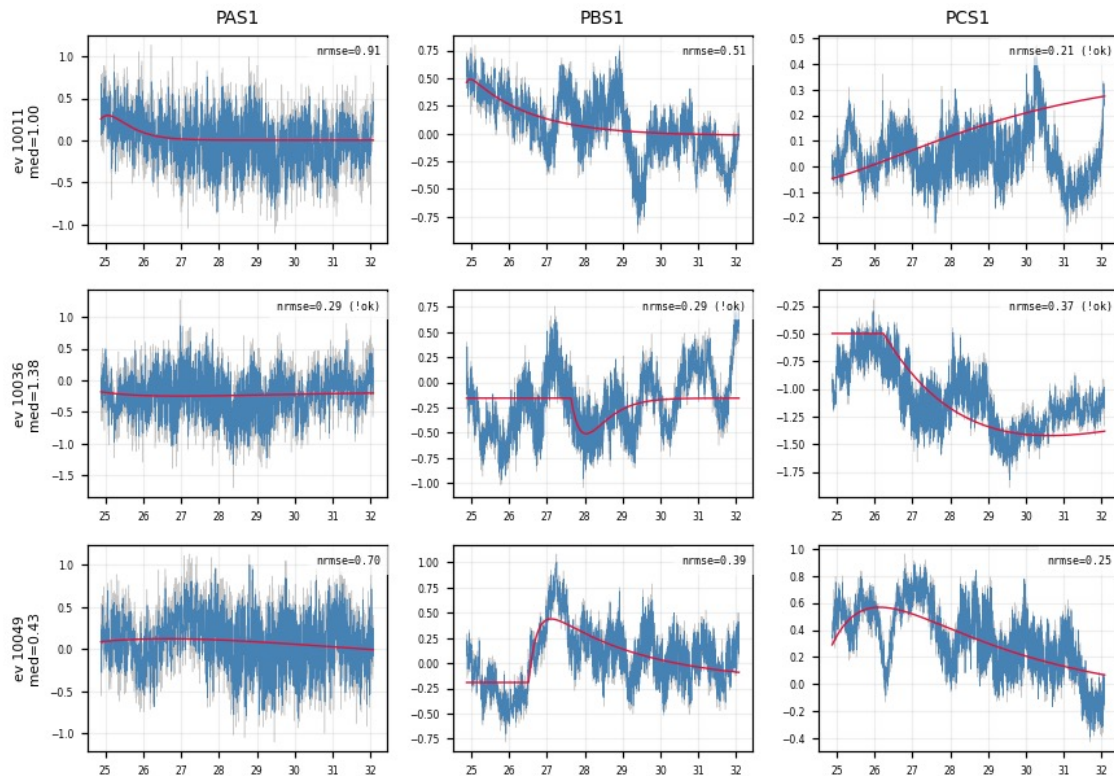
Z7 PBS1 (quiet): two clean populations, log-log axes



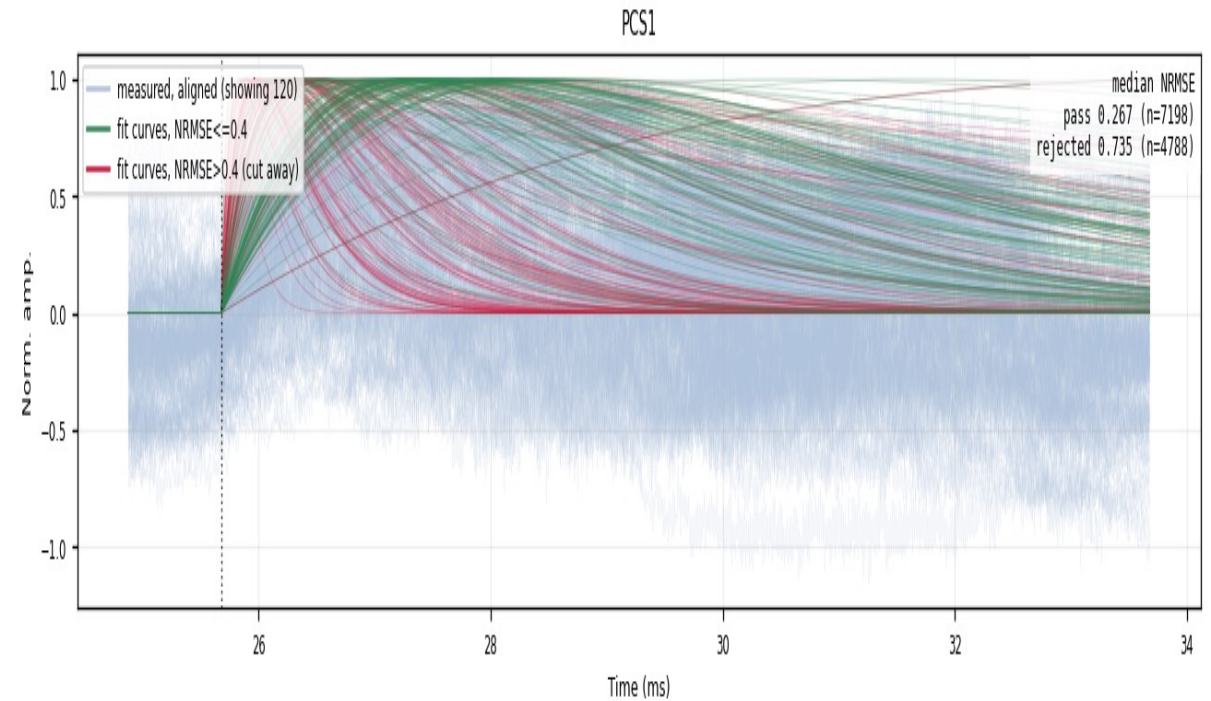
Z22 PAS1 (weak): noise population dominates — the window alone is not enough

# The rejected population is noise — verified in the raw traces

- Event grids of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse — the cut removes noise triggers, not physics
- Fan-cut view: fitted curves that pass (green) vs cut away (red) on top of the aligned data, median NRMSE of each population stamped per channel



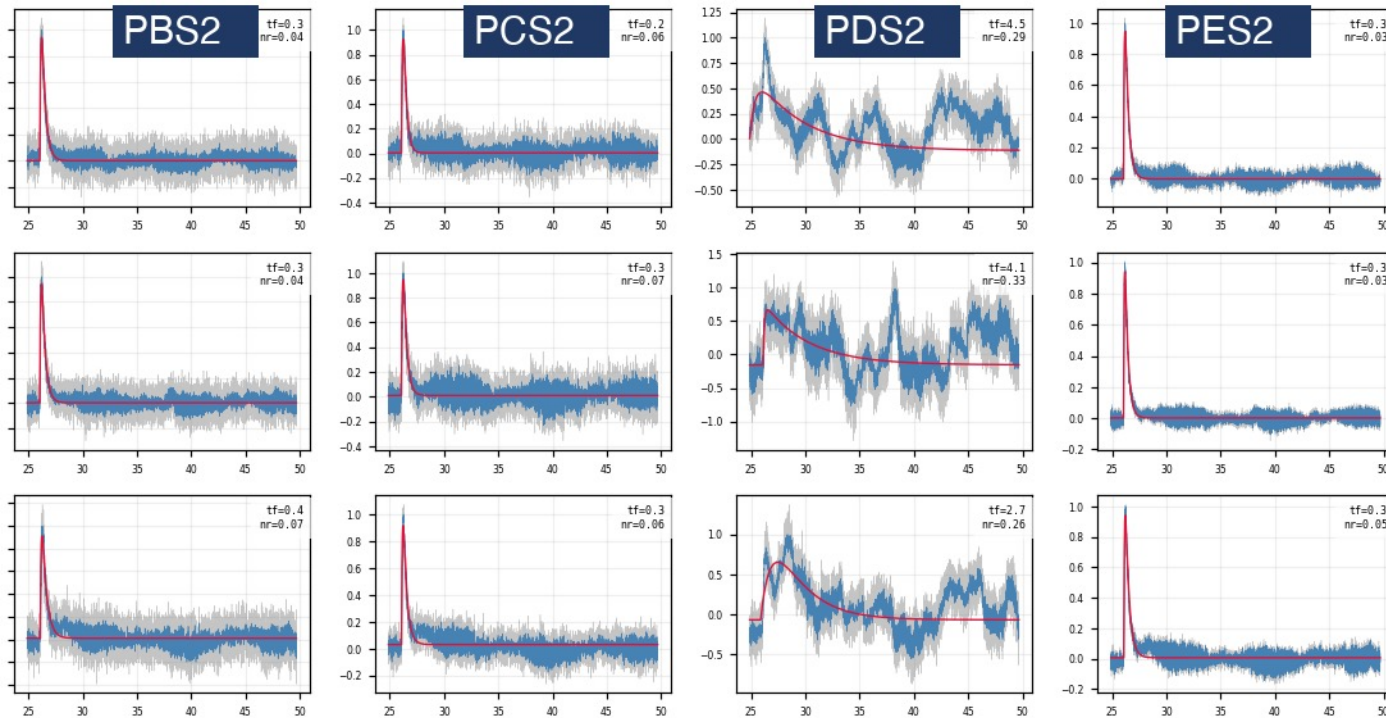
Z22: NRMSE-rejected events — raw traces are noise



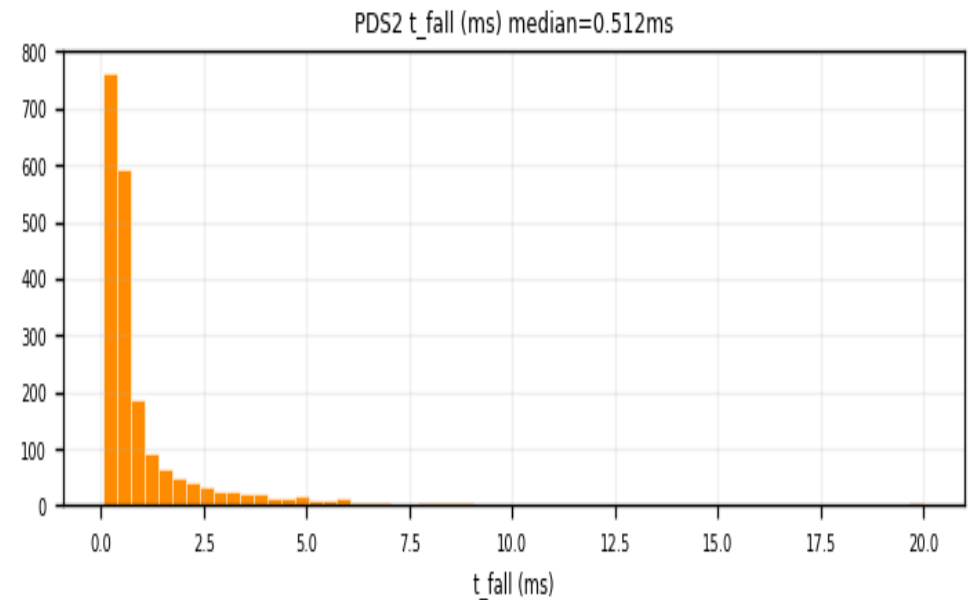
Z22 PCS1: pass (green) vs rejected (red) — vertically stretched for visibility

# Follow-up 1 — the slow-fall tail is a one-channel artifact

- The post-cut fan still shows slow-fall tails → sample them:  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{fall}} > 1.5$  ms, 10 random events, raw vs fit drawn in all 12 channels
- The sampled events are real pulses → kept. Their extreme fall times trace to one channel: only PDS2 swings ( $\tau_{\text{fall}}$  median 0.51 ms vs  $\approx 0.25$  ms elsewhere) — a low-frequency disturbance
- Conclusion: no  $\tau_{\text{fall}}$  cut — slow-fall events passing NRMSE stay in; the extreme tail is a one-channel artifact, not physics



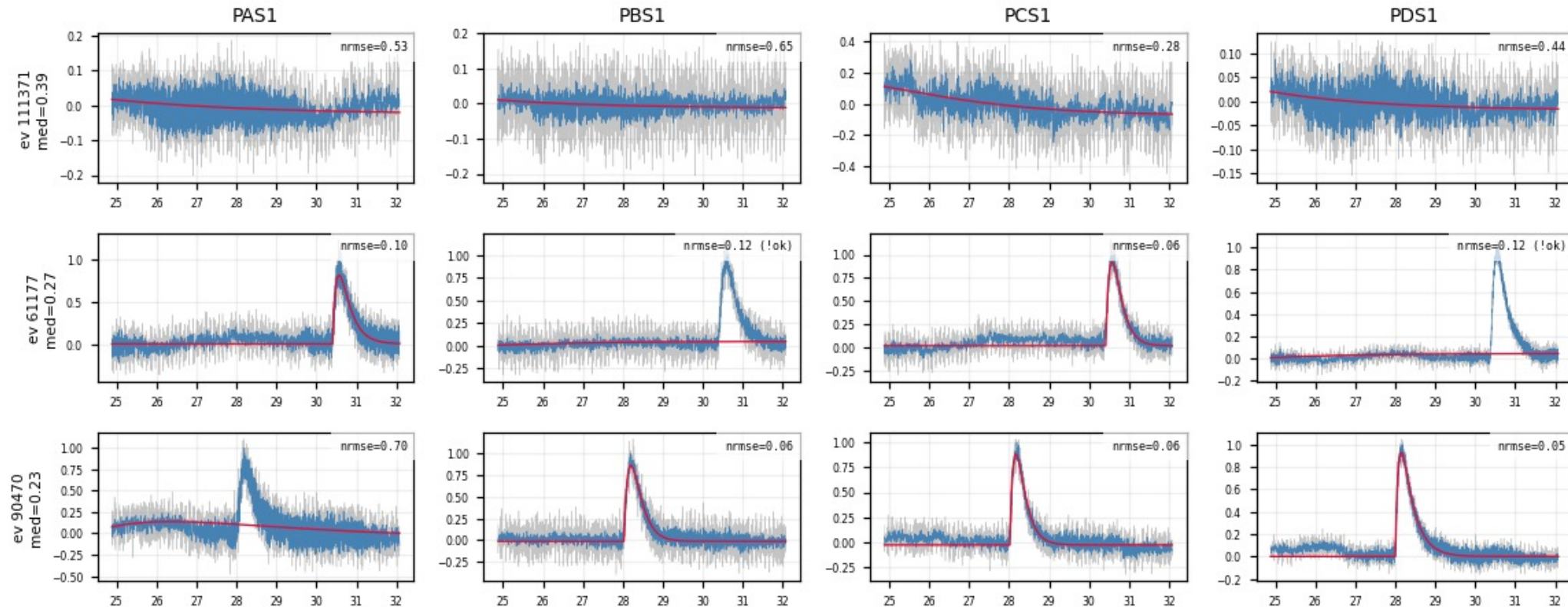
Z7, 3 of the sampled slow-fall events: normal pulses in PBS2/PCS2/PES2, low-frequency swings only in PDS2



Z7 PDS2, fitted  $\tau_{\text{fall}}$ : broad tail (median 0.51 ms) — the artifact in the distributions

## Follow-up 2 — genuine slow-rise pulses (“shadow” events)

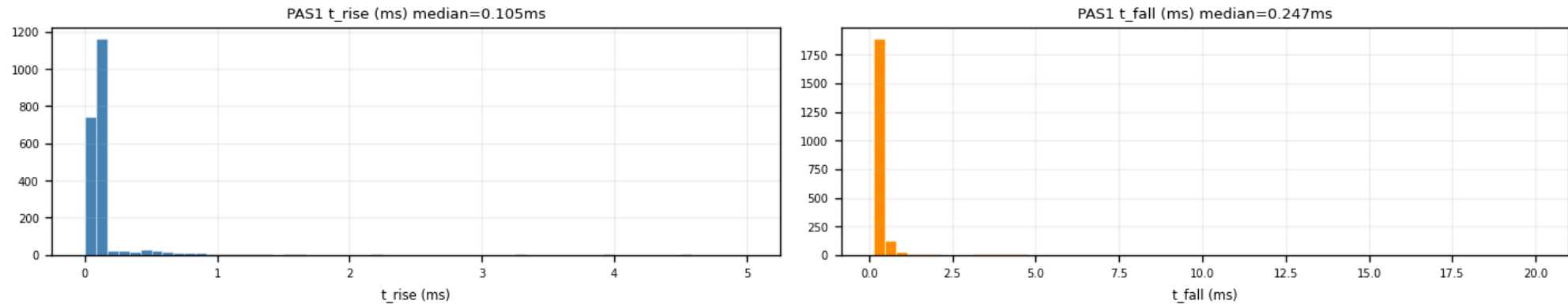
- Same sampling recipe on the rise side: median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms  $\rightarrow$  the raw traces show real pulses — onset aligned, peak late, consistent across channels
- They are the faint displaced bundle in the aligned overlays — a genuine second pulse shape (candidate surface/bulk effect, not settled)
- Real physics  $\rightarrow$  cannot simply be cut away; exactly the shape variation the multi-template NxM method is built to capture



Z7 shadow events (first 4 channels): raw (gray) / LP (blue) / fit (red) — genuine slow pulses, onset aligned, peak late

# Slow drift also fits “slow”: a $\tau_{\text{rise}} \leq 0.3$ ms ceiling

- On noisy-window detectors a smooth baseline drift survives the NRMSE cut — a slow 2-exp hugs it with a tiny residual — and mimics a slow rise
- Fast pulses sit at  $\tau_{\text{rise}} \approx 0.1$  ms (p90  $\approx 0.15$  ms), far below the drift tail  $\rightarrow$  a ceiling at 0.3 ms blocks the drift at  $\approx 2\text{--}5\%$  signal cost
- Keeps most genuine slow pulses, trims the very slowest — documented trade-off, final decision pending

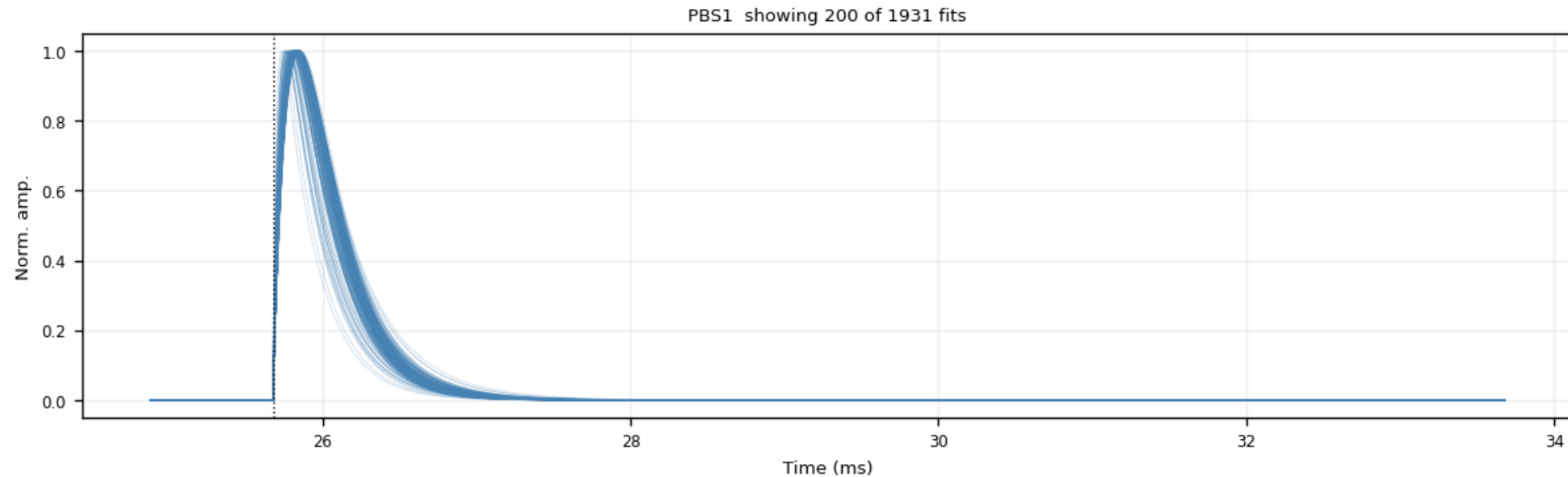


*Z7 PAS1: fitted  $\tau_{\text{rise}}$  (median 0.105 ms) and  $\tau_{\text{fall}}$  distributions*

- On quiet detectors both  $\tau_{\text{rise}}$  and  $\tau_{\text{fall}}$  distributions are narrow — the pulse shape is stable across events

# Template family 1 — analytic 2-exp, NRMSE-weighted (1x1)

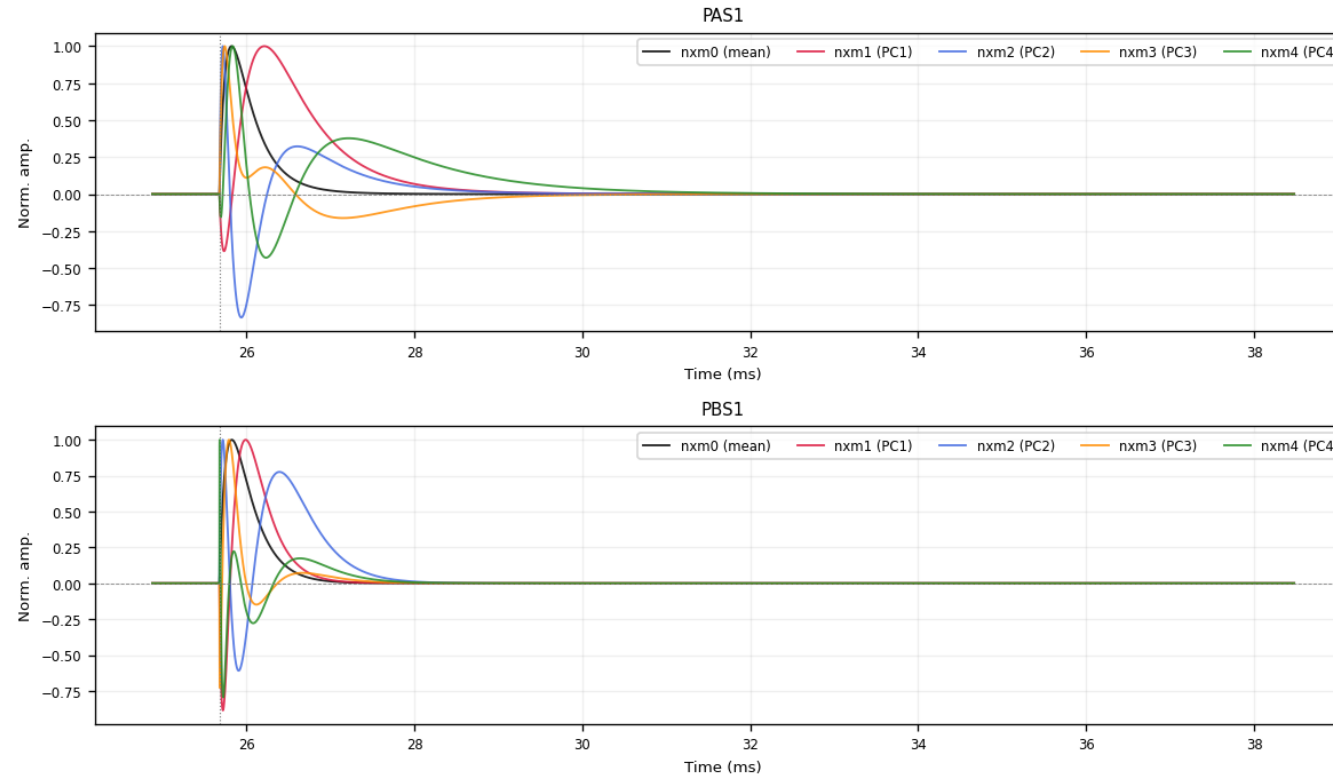
- Per channel: weighted mean of ALL fit\_ok fitted curves at the common onset, weight  $w = 1/\max(\text{NRMSE}, 0.01)^2$
- Badly-fit events count less but are never excluded — robust and smooth (noise-free by construction)
- Delivered as peak-normalized 32768-bin ROOT TH1D per channel + summed PT/PS1/PS2 templates



*Z7 PBS1, fitted curves after  $\text{NRMSE} \leq 0.4$  and  $\tau_{\text{rise}} \leq 0.3$  ms — the NRMSE-weighted mean of this family is the 1x1 template*

# Template family 2 — NxM PCA templates

- Input: fitted curves passing  $\text{fit\_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3$  ms, at common onset, peak-normalized (PCA window 15550–24050,  $\leq 3000$  curves/channel)
- $\text{nxm0}$  = mean shape;  $\text{nxm1} \dots \text{nxm4}$  = first four principal components — a real pulse is fit as  $\sum_i \text{amp}_i \cdot \text{nxm}_i$
- PC1 + PC2 already capture 96–98% of the shape variance; final step before delivery: normalize all five to unit peak — the delivered product



Z7 PAS1 / PBS1:  $\text{nxm0}$  (mean) +  $\text{nxm1} \text{--} 4$  (PCs) — the oscillating components encode rise/fall-time variation

# Delivered — and what remains

---

- Delivered for all 13 zips, official cdmsbats PulseTemplates layout (zip{N}/{chan}, {chan}nxm0–4, summed PT/PS1/PS2):
  - SNOLAB\_R4\_20260706\_ZhihengLi\_zip{N}.root — 2-exp NRMSE-weighted (1x1)
  - SNOLAB\_R4\_20260707\_ZhihengLi\_pca\_zip{N}.root — NxM PCA (normalized)
- Every step is traceable: raw cache → per-series fit checkpoints indexed by EventNumber → self-documenting figures (11 types × 13 zips, all versioned)
- Cuts derived from the data, validated in raw traces:  $\text{NRMSE} \leq 0.4$  (bimodal valley),  $\tau_{\text{rise}} \leq 0.3$  ms (drift leakage)
- Open items:
  - final decision on the  $\tau_{\text{rise}}$  ceiling vs the genuine slow-pulse population
  - run the group's template-validation method on the new templates